VIT
Vellore Institute of Technology
CHENNAI

Reg. Number: 19

Continuous Assessment Test (CAT) – I - AUG 2024

Programme	:	B.Tech Computer Science and Engineering / B.Tech. CSE (CPS)	Semester	:	Fall 24-2025
Course Code & Course Title	:	BCSE415L - Human Computer Interaction	Class Numbers	:	CH2024250100654 CH2024250101755 CH2024250101754 CH2024250101759 CH2024250101760 CH2024250101757
Faculty(s)	:	Dr. Deepika Roselind J Dr. Sandosh Dr. Ancy Micheal A Dr. Premanand V Dr. Arthi M Dr. Sudha C	Slot	:	E1+TE1
Duration	:	1 hour 30 Minutes	Max. Mark		50

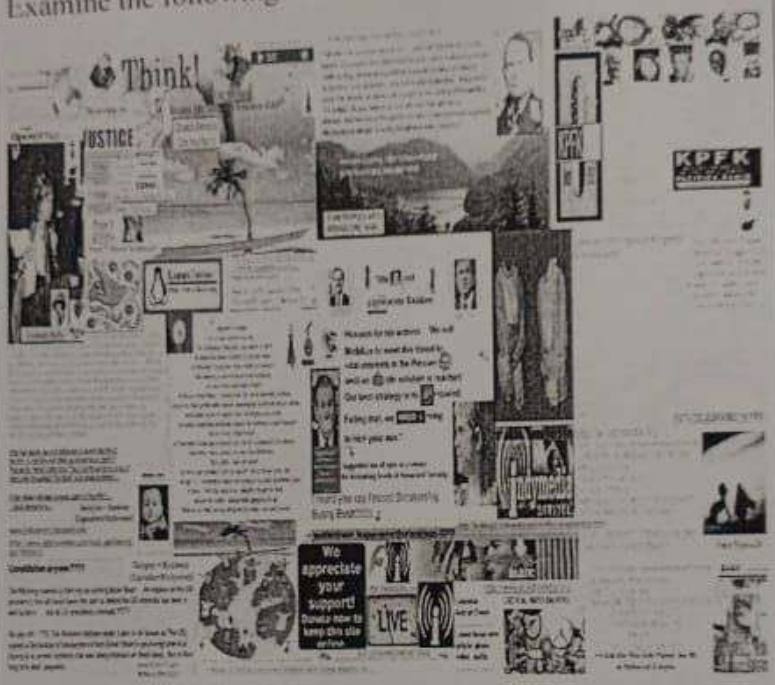
General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information.

Answer all questions

Q. No	Sub Sec.	Description	Marks
1		Imagine that you are designing a smart assistant for a new healthcare application that helps patients track their symptoms, schedule appointments, and receive personalized health tips. The smart assistant must also be able to suggest potential diagnoses based on user-reported symptoms and recommend whether they should visit a doctor immediately, schedule a check-up, or monitor their condition at home. Design a UI and explain in detail on how you would apply reasoning techniques to improve the decision-making capabilities of the smart assistant when a person is suffering from diabetes.	10

2		Assume that you are tasked with setting up a new computer lab in your university. The lab must be designed to support a wide range of activities such as coding, graphic design, virtual reality (VR) and video conferencing. i. Identify and select the appropriate input and output devices for the computer lab to ensure it meets the needs of coding, graphic design, VR, and video conferencing? [4 M] ii. Propose strategies for implementing customizations in the new computer lab to effectively accommodate disabled users. Elaborate with a suitable example. [6 M]	10
3		Ajay is a UX designer working on a food ordering app that offers a variety of menu options for users to browse and select food from various restaurants. His goal is to design the menu. He has to consider various factors such as the number of clicks or taps required to access different menu sections, the cognitive load associated with menu organization, item selection and the sequence of user actions from menu browsing to placing an order. Design UI and compare using GOMS to optimize the sequence of steps required for efficient 'menu design' and 'order placing' scenario for a novice and an experienced user.	10
4		Sarah is designing a mobile application for a travel planning service. The app includes some core features such as Home, Search Destinations, My Trips, Booking History and Profile Settings. Each feature needs to be easily accessible from anywhere in the app. i. Illustrate any two suitable navigation structures that will allow users to quickly move between these sections without excessive clicks or confusion. [7 M] ii. Discuss any design considerations you would make to ensure that the navigation is intuitive and efficient for users. [3 M]	10

5		Examine the following interface:  Specify the UI Golden Rules that have been violated by this interface. Examine the impact of adhering to the Golden Rules on increasing the user productivity.	10

**********All the best **********